

Nicole Ye

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TECHNICAL SKILLS

Firmware:: ESP32, Raspberry Pi Pico, C++, Java, PlatformIO, I2C, SPI, UART, VL53L4CX, RDM630 RFID

Electronics: KiCad, EasyEDA, PCB schematic design, Soldering, Cable Management, Crimping

Mechanical/CAD: Onshape, SolidWorks, BambuStudio, 3-D Printing, Snap-Fits

Tools: VSCode, Multimeter, TKO Circuit Simulator, Excel QA tracking

EXPERIENCE

Prometheus Industries (GreaseTech) — Mechatronics Engineering Intern Jan. 2026 – Apr. 2026

- Integrated **VL53L4CX** time-of-flight sensor via **I2C on ESP32** for threshold-based canister detection, replacing an unreliable optical sensor and eliminating false-positive reads in production firmware.
- Developed hall sensor firmware (C++) for rotational grease-dispense tracking across 14+ assembled units.
- Resolved RFID read inconsistency across 200+ sorted tags by characterizing per-type antenna offsets and iterating nozzle grip geometry in **Onshape** to a unified offset reliable for all tag variants.
- Designed custom buck converter schematic around SKMW15F-05 to replace off-the-shelf module, targeting 40%+ footprint reduction; executed full PCB bring-up including JST connectors, RDM630 RFID reader, and wiring.
- Iterated electronics enclosure system across 3 CAD revisions in **Onshape/SolidWorks**, reducing volume ~30% and integrating wire channels, PCB standoffs, and heat-insert mounts.
- Validated 8 nozzle geometry variants via 3D-printed QA testing, determining 84 mm from grip end as optimal RFID antenna reference point; authored manufacturing assembly documentation with annotated diagrams.

Team 2198 FIRST Robotics Team — Captain/Electrical Lead Sep. 2021 – Jun. 2025

- Captained a 30+ member team to design, build, and compete with a full-scale FRC robot across **4 seasons**, achieving a **top-10 regional finish** in 2024–2025
- Engineered electrical system using [TKO Circuit Simulator](#), integrating colour sensors, magnetic sensors, and encoders for real-time game-piece detection **10+ electrical components**
- Programmed AprilTag-based field-relative localization in **Java** using Limelight vision co-processor, enabling automated robot alignment to scoring positions and reducing driver repositioning time during matches
- Designed in **Onshape** 5+ sub-mechanisms (elevator, intake, climber, shooter), coordinating design-to-manufacture handoff within a **6-week** build season
- Mentored **15+ junior members** in fabrication — drill presses, bandsaws, and power tools — enabling independent assembly of sub-mechanisms
- Produced a [team website](#) and [sponsorship package](#) that secured **\$6,000+** in funding for the 2024–2025 season

UW Robotics Mars Rover Design Team Electrical/Firmware Member Sep. 2025 - Present

- Designed custom PCB for a 555 timer-based circuit, including schematic layout using DipTrace
- Documented the PCB design process and created clear schematics for future reference or replication
- Developed a CAN bus communication protocol using C++ to control servo motors using PWM signals

PROJECTS

Tutoring Service - Space Academy | *JavaScript, HTML, CSS*

- Created [event-driven UI](#) components with JavaScript, simplifying course selection and enhancing user interaction
- Implemented dynamic shopping cart functionality using JavaScript, including real-time updates, item management, and automatic total cost calculations
- Created interactive buttons to select courses, add them to a shopping cart, and proceed to checkout

AWARDS

Queen's Chancellor Scholarship Receipt (\$48,000), University of Toronto Book Scholar Award

L'Amoureux C.I's Student of the Year, Innovation in Control Award - ONT First Robotics 2024 Humber College Event

EDUCATION

University of Waterloo

Honour's Electrical Engineering, Bachelor's of Applied Science

Sep. 2025 - Apr. 2030

Waterloo, ON, Canada